Engineering the microbiome under individualized perturbations

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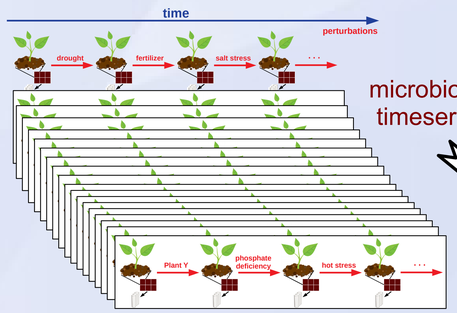
- Motivation: **Microbiome dynamics** studies highlight the inability to predict the effects of external perturbation on complex microbial communities over time
- Methods: MDPbiome is an Artificial Intelligence system built using **Markov Decision Processes (MDP)**, that provides in-silico recommendations (e.g., about diet, pre/pro-biotics, drugs, etc.) to guide the subject's microbiome through a path towards health or high performance. MMODES is a tool to simulate the dynamics of microbial communities under perturbations using genome-scale metabolic models (GEMs), given a lack of experimental longitudinal microbiome data. MDPbiome plus MMODES = MDPbiomeGEM
- Results: MDPbiome and MDPbiomeGEM systems have been successfully **applied to several real and simulated case studies** in human, animal and soil microbiomes, to chick gut flora maturation, soil decontamination, to recover from Crohn's disease, and to avoid bacterial vaginosis

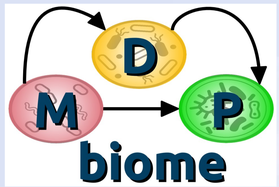
Approach

MDPbiome [García-Jiménez et al., Bioinformatics, 2018]

Experimental data



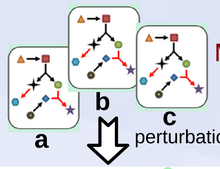
microbiome timeseries



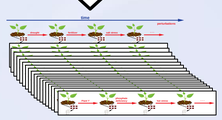
MDPbiome concepts (I):
- Markov Decision Process (MDP)
- Microbiome states

MDPbiomeGEM [García-Jiménez et al., pre-print, 2020]

Simulated data

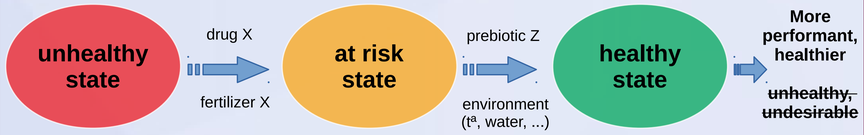


GEnome-scale Metabolic models (GEM)



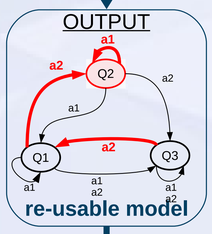
MMODES:
- Combines microbiome and metabolism
- Simulates microbial community over time (biomass and medium composition)

Goal



Microbiome engineering: guiding microbiome dynamics with perturbations

Individual



Individualized intervention

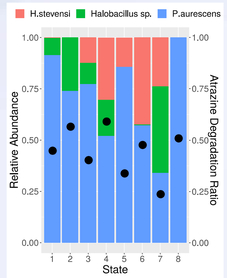
MDPbiome concepts (II):
- **Flexible goal** state definition to optimize (red ellipse)
- **State-transition diagram**, labeled by perturbation frequency

Predictions consistent with biomedical and environmental literature

We are looking for collaboration for experimental validation!!

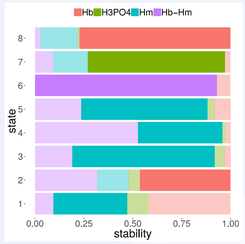
Results

Case study: Herbicide degradation
(simulated data: ~100 time series)

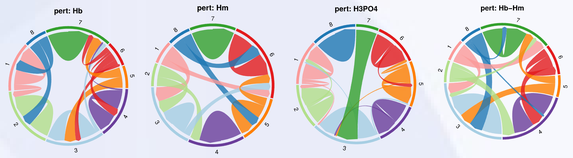


Microbiome states
Microbial composition and optimization value of distinct identified states

Assessment of the recommendation, highly stable recommendations to maximize atrazine degradation



Transition diagrams
Explaining dynamics depending on the perturbation



Contributions:
- Address challenge of moving microbiome studies from descriptive to translational approaches
- Generate predictive individualized intervention plans
- Empowering microbiome engineering, by data-driven hypotheses
- High flexibility: applicable to human, animals, plants, synthetic communities, etc; to experimental or simulated datasets

Do you want to engineer/predict your microbial community with AI? Let's us know!!

@beagj

http://wilkinsonlab.info/microbiomes

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https://github.com/beatrizgj/MDPbiome
https://github.com/carrascomj/mmodes
https://github.com/beatrizgj/MDPbiomeGEM